BURNABY, BC, CANADA

Mohammad Shahidzadeh Asadi

Computer Engineering Researcher · FPGA Systems · RISC-V · Hardware/Software Co-design

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PROFILE

Computer engineering researcher working across FPGA-based heterogeneous systems, RISC-V processors, operating systems, hardware verification, and FPGA CAD. Experienced in taking ideas from algorithms and RTL through Linux bring-up, FPGA implementation, and empirical evaluation.

EDUCATION

M.A.Sc. in Computer Engineering 2023-2026

Simon Fraser University · Burnaby, Canada

Thesis: *Operating System Support for Accelerators as First-Class Compute Elements in Heterogeneous Systems*. Advisor: Prof. Lesley Shannon.

B.Sc. in Computer Engineering 2018-2023

Shahid Bahonar University of Kerman · Iran

GPA: 3.99/4.00. Top-three GPA rank among 120 computer engineering students.

RESEARCH EXPERIENCE

Research Assistant 2023-2026

Reconfigurable Computing Lab · Simon Fraser University

Developed Linux-capable FPGA systems, RISC-V processor extensions, hardware/software mechanisms for accelerator support, and LLM-assisted formal-verification tooling.

- Built MEERKAT, a configurable CVA5-based platform for measuring and reducing accelerator communication, invocation, interrupt, and scheduling overheads.
- Extended CVA5 and its SoC environment with Linux support including privilege, virtual memory, PLIC/CLINT interrupts, OpenSBI, and LiteX integration.
- Created a fine-tuned, iterative LLM flow for SystemVerilog assertion generation; published at ISQED 2025.

Research Assistant 2021-2023

Shahid Bahonar University of Kerman

Contributed to FPGA CAD, dependable DNNs, FPGA accelerator tooling, and GPU-accelerated logic resimulation.

- Modified VTR/VPR for congestion-aware placement and investigated hMETIS-based hypergraph packing and placement.
- Extended a DNN accelerator builder with additional layers, a PyTorch workflow, and LiteX SoC integration.
- Co-developed the first-place solution for ICCAD 2020 Problem C: GPU-Accelerated Logic Resimulation.

PEER-REVIEWED PUBLICATIONS

Mohammad Shahidzadeh, B. Ghavami, S. J. E. Wilton, and L. Shannon. “Automated Verilog Assertion Generation Using Fine-Tuned LLMs with Subtask-Specific Iterative Prompting.” *ISQED*, pp. 1-7, 2025.

B. Ghavami*, **Mohammad Shahidzadeh***, L. Shannon, and S. J. E. Wilton. “ZOBNN: Zero-Overhead Dependable Design of Binary Neural Networks with Deliberately Quantized Parameters.” *IOLTS*, 2024. *Equal contribution.

B. Ghavami, M. Sadati, **Mohammad Shahidzadeh**, L. Shannon, and S. J. E. Wilton. “A Semi Black-Box Adversarial Bit-Flip Attack with Limited DNN Model Information.” *ICCD*, pp. 96-104, 2024.

B. Ghavami, M. Sadati, **Mohammad Shahidzadeh**, Z. Fang, and L. Shannon. “Blind Data Adversarial Bit-Flip Attack against Deep Neural Networks.” *DSD*, pp. 899-904, 2022.

SELECTED RESEARCH PROJECTS

MEERKAT: Accelerators as First-Class Compute Elements 2024-2026
M.A.Sc. thesis · HPCA 2027 submission under review
Linux-capable FPGA framework that moves selected O/S support into hardware. Achieved up to 260x lower communication overhead, up to 71x faster hot-page translation, and 1.5-5.8x faster wall-clock scheduling in evaluated configurations.

Linux Support for CVA5 2023-2025
RISC-V processor and SoC extension
Implemented and debugged architectural support for Linux, including PLIC, CLINT, privilege modes, virtual memory, kernel-required instruction behavior, OpenSBI boot, and LiteX-based FPGA systems.

Automated Verilog Assertion Generation 2023-2025
Formal verification with fine-tuned LLMs
Developed datasets, subtask-focused fine-tuning, compiler feedback, and iterative repair. Improved functionally correct assertions by 7.3x and syntax-error-free assertions by 26%.

DNN Accelerator Builder 2022-2023
Full-stack model-to-FPGA flow
Extended layer support, built a PyTorch-facing conversion path, developed streaming RTL, and integrated the accelerator into the LiteX SoC flow.

Congestion-Aware FPGA CAD 2021-2023
VTR/VPR placement and hMETIS packing
Added congestion information to placement costs and explored hypergraph-based packing and placement. Public work is documented in VTR pull request #2384.

GPU-Accelerated Logic Resimulation 2020
ICCAD 2020 CAD Contest · Problem C
Developed a Verilog-to-C++ netlist compiler and delay-aware CUDA resimulation engine. First place among 20 teams.

HONORS AND AWARDS

First Place · CAD Contest at ICCAD 2020
Problem C: GPU-Accelerated Logic Resimulation · Team cada0143
Official contest winner with Seyed Mani Sadati; advised by Prof. Behnam Ghavami.

Bronze Medal · ACM-ICPC Asia Tehran Regional 2018/19
Regional rank 4; online contest rank 1 among 58 teams
Competitive programming recognition.

Top-Three GPA Rank 2018-2022
Shahid Bahonar University of Kerman
Ranked in the top three among 120 computer engineering students.

TECHNICAL SKILLS

HARDWARE	SystemVerilog, Verilog, VHDL, RTL design, FPGA implementation, RISC-V, DMA, PLIC, CLINT
FPGA / CAD	Vivado, VTR, VPR, Yosys, Quartus, LiteX, hMETIS, packing, placement, routing
VERIFICATION	JasperGold, SymbiYosys, Verilator, ModelSim, SystemVerilog Assertions, formal property verification
SOFTWARE	C/C++, Python, CUDA, Linux kernel, OpenSBI, PyTorch, Docker, Git, GDB, Valgrind
PLATFORMS	Xilinx VCU118, Alveo U200, ZedBoard, Arty A7, Cyclone V, BeagleBone, Raspberry Pi

RESEARCH IN REVIEW

Operating System Support for Accelerators as First-Class Compute Elements in Heterogeneous Systems. Submitted to HPCA 2027; decision pending. This item is listed separately from peer-reviewed publications.